# Girls Build Bots — Brownie Automotive Engineering

## Badge-aligned workshop guide for robotics-team-led outreach

**Level:** Brownies (Grades 2–3) · **Badge:** Automotive Engineering — Automotive Engineering **Series:** Automotive Engineering (Design · Engineering · Manufacturing) **Format:** One 75–90 minute session · **Ratio:** 1 mentor per 3–4 Scouts **Who runs it:** A FIRST robotics team (FLL / FTC / FRC) leading the session. A physical robot is a nice bonus but **not required** — every activity runs on craft and cardboard supplies the team or troop can provide.

> **Troop Leaders:** This curriculum is designed to *align with* the Brownie Automotive Engineering badge series. Official badge requirements are in your Girl Scout Volunteer Toolkit (VTK). Verify completed activities against the official steps before awarding badges. Badges are purchased through your council shop.

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## The through-line: designing for a driver

The Automotive series follows a car from idea to road: design it, **engineer** it to move, then manufacture it. Across all three, Brownies learn that engineers build for a *person* — and that engineers **test and improve** their work.

**Vocabulary:** engineer, force, ramp, test, improve, friction (grippy vs. slippery), fair test.

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## Automotive Engineering (75–90 minutes)

**Badge theme:** Engineering a vehicle that actually moves — building it, testing it on a track, and improving the design based on what happens.

### Materials

- Cardboard, bottle caps or plastic lids (wheels), skewers or straws (axles), tape
- A ramp (a board or stiff cardboard propped on books/chairs)
- Tape measure or a floor marked with tape lines; sticky notes for recording distances

### Activity 1 — What Makes a Car Go? (15 min)

No engine today — we use a ramp and **gravity**. Talk about **force** (a push or pull) and **friction** (grippy slows you down, slippery lets you roll). Quick prediction: will a heavier car or a lighter car roll farther? Everyone guesses — engineers predict before they test.

### Activity 2 — Build Your Roller (25 min)

Scouts build a simple cardboard car with axles and wheels that spin freely. Mentors help troubleshoot wobbly wheels (a real engineering problem!). Goal: it rolls straight down the ramp.

### Activity 3 — Test, Change, Test Again (30 min)

The engineering heart of the session. Each car rolls down the ramp; Scouts measure how far it goes and record it. Then change **one thing** — add weight, swap wheels, straighten the axle — and test again. Did it help? **The rule: change only one thing at a time**, or you won't know what worked. That's a *fair test*.

### Activity 4 — Best Improvement Wins (5 min) + Wrap (5 min)

Not "whose car is fastest," but "who made the biggest *improvement*." Each Scout shares one change she made and what happened. Celebrate the engineering, not just the winner.

> **Robot moment (optional):** Share for 60 seconds how the team tests the robot over and over, changing one thing at a time, to make it work better — the same fair-test habit.

### Completion Checklist — Automotive Engineering

By the end of the session, to earn the badge, each Scout can: - [ ] Explain that a force (like gravity on a ramp) makes a car move - [ ] Build a car with wheels that roll - [ ] Test her car and measure how it did - [ ] Change one thing, test again, and say whether it improved

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## Mentor Guide (for the robotics team)

- **You don't need a robot for this.** A ramp and cardboard cars are the whole show. The "robot moment" box is optional.
- **One variable at a time.** This is *the* lesson — enforce it kindly. Changing two things at once is the teachable "mistake."
- **Predict before every test.** Ask "what do you think will happen?" each round.
- **Reframe winning.** Celebrate the biggest improvement, not the fastest car, so every Scout can succeed.
- **Wobbly wheels are a gift.** Troubleshooting them together is real engineering — don't just fix it for her.

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## 📣 Share Your Impact — required for teams and troops using this curriculum

If you use Girls Build Bots materials to run a workshop, please complete all three:

1. **Credit us.** Tell your Scouts and audience that the curriculum comes from Girls Build Bots, and credit "Girls Build Bots" anywhere you share about the event.
2. **Tag us.** Post about your workshop and tag **@girlsbuildbots** on Instagram.
3. **Report your results.** Email **girlsbuildbots@gmail.com** with: the number of girls who participated, the number of badges earned, and a sentence or two of feedback on how it went.

This is how the program grows and how we measure our impact — your report becomes part of the Girls Build Bots story and helps the next troop and team.

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*Created for the "Girls Build Bots" Girl Scout Gold Award project. Original curriculum aligned to the Brownie Automotive Engineering badge series; official requirements © Girl Scouts of the USA, available to troop leaders in the Volunteer Toolkit.*